EDS 223: Homework Assignment 1

Toxic Releases to Air and Air Toxics Cancer Risk in Southwestern San Bernardino County, California

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Toxic Releases to Air and Air Toxics Cancer Risk in Southwestern San Bernardino County, California

Introduction

San Bernardino County, California is the state's largest county by total land area *(5)*, with the majority of its cities located in the southwesternmost 2% *(4)* of the county. According to the American Lung Association State of the Air 2026 report's assessment of air quality montoring data from the United States Environmental Protection Agency (US EPA), San Bernardino County is ranked the number one most polluted county for ozone and particle pollution in the United States *(1)*.

In addition to air quality montioring, the US EPA tracks the release of hundreds of toxic chemicals through the Toxic Release Inventory (TRI) program *(3)*, including chemicals subject to reporting under the Clean Air Act. In an area known to be impacted by pollution, investigating air quality impacts beyond ozone and particle pollution may reveal addional potential health impacts worth investigating in the region. The following maps were created to explore toxic releases to air and air toxics cancer risk in southwestern San Bernardino County.

Data for these maps were sourced from the US EPA's previous EJScreen: Environmental Justice Screening and Mapping Tool.

Load Packages

```
# Load packages
library(tidyverse)
library(sf)
library(tmap)
library(here)
```

Import Data

```
# Read in geodatabase of EJScreen data at the Census Block Group level
ejscreen                    <-                    sf::st_read(here::here("data",
"ejscreen","EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.gdb"))
```

```
Reading layer `EJSCREEN_StatePctiles_with_AS_CNMI_GU_VI' from data source
                                                                        `C:
\Users\prisc\Documents\UCSB\MEDS\EDS_223_Geospatial_and_Remote_Sensing\eds223-
hwk1\data\ejscreen\EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.gdb'
  using driver `OpenFileGDB'
Simple feature collection with 243021 features and 223 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -19951910 ymin: -1617130 xmax: 16259830 ymax: 11554350
Projected CRS: WGS 84 / Pseudo-Mercator
```

```
# Filter to California
california <- ejscreen |>
  dplyr::filter(ST_ABBREV == "CA")

# Filter to San Bernardino County
san_bernardino <- ejscreen |>
  dplyr::filter(CNTY_NAME %in% c("San Bernardino County"))
```

Visualize Data

The following maps explore toxic releases to air and air toxics cancer risk in southwestern San Bernardino County.

The **toxic releases to air indicator** was determined by US EPA EJScreen using data reported by facilities to the US EPA's Toxic Release Inventory (TRI) program through 2021 Risk-Screening Environmental Indicators modeled results. The TRI program covers chemicals that are known to cause cancer, significant adverse chronic and acute human health effects, and signficant adverse environmental effects *(3)*. This indicator is intended to visualize modeled toxicity-weighted concentrations of TRI-listed chemicals in the air.

The **air toxics cancer risk indicator** was determined by US EPA EJScreen using 2019 US EPA Air Quality Planning and Standards data. This indicator is intended to visualize lifetime cancer risk from inhalation of air toxics.

Statistical percentile data for the two indicator values were used in order to represent the indicators clearly for comparison.

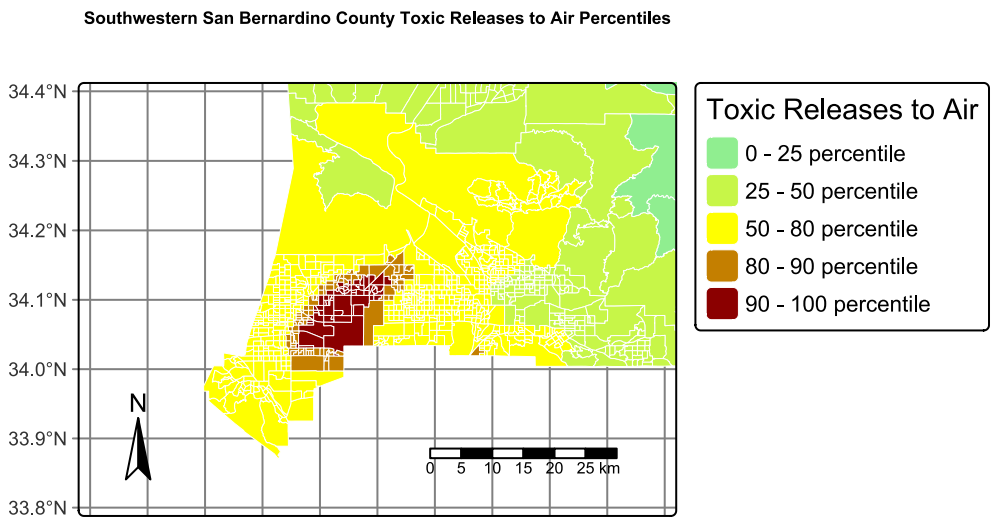
Toxic Releases to Air

```
tmap_mode('plot')

# Visualize toxic releases to air percentiles

sw_sb_toxic_release <- tm_shape(san_bernardino, bbox = c(-118, 33.8, -117, 34.4))
+ # Add bounding box to restrict to southwestern area of county
    tm_graticules() + # Add lat and long lines
    tm_polygons(fill = 'P_RSEI_AIR', # Add toxic releases to air percentile data
                fill.scale = tm_scale_intervals(
                  breaks = c(0, 25, 50, 80, 90, 100),
                  values = c('lightgreen', 'yellow', 'darkred')),
                fill.legend = tm_legend(
                  title = 'Toxic Releases to Air',
                   labels = c('0 - 25 percentile', '25 - 50 percentile', '50 -
80 percentile', '80 - 90 percentile', '90 - 100 percentile'),
                  position = tm_pos_out('right')),
                col = 'white',
                lwd = 0.2) +
    tm_compass(position = c('left', 'bottom'), # Add compass
               size = 2) +
    tm_scalebar(position = c('right', 'bottom')) + # Add scalebar
       tm_title('Southwestern San Bernardino County Toxic Releases to Air
Percentiles',
              fontface = 'bold')

sw_sb_toxic_release
```

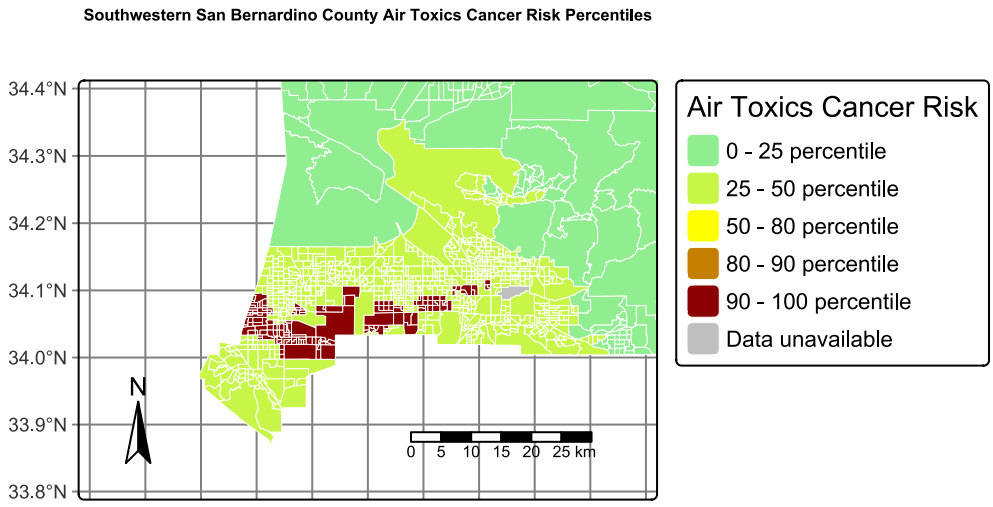


Air Toxics Cancer Risk

```
# Visualize lifetime cancer risk from inhalation of air toxics

sw_sb_cancer_risk <- tm_shape(san_bernardino, bbox = c(-118, 33.8, -117, 34.4))
+ # Add bounding box to restrict to southwestern area of county
    tm_graticules() + # Add lat and long lines
    tm_polygons(fill = 'P_CANCER', # Add air toxics cancer risk percentile data
                fill.scale = tm_scale_intervals(
                  breaks = c(0, 25, 50, 80, 90, 100),
                  values = c('lightgreen', 'yellow', 'darkred')),
                  fill.legend = tm_legend(
                    title = 'Air Toxics Cancer Risk',
                  labels = c('0 - 25 percentile', '25 - 50 percentile', '50 - 80
percentile', '80 - 90 percentile', '90 - 100 percentile', 'Data unavailable'),
                    position = tm_pos_out('right')),
                  col = 'white',
                 lwd = 0.2) +
    tm_compass(position = c('left', 'bottom'), # Add compass
               size = 2) +
    tm_scalebar(position = c('right', 'bottom')) + # Add scalebar
      tm_title('Southwestern San Bernardino County Air Toxics Cancer Risk
Percentiles',
              fontface = 'bold')
```

```
sw_sb_cancer_risk
```



Interpretation

Toxic releases to the air are concentrated to the south in a distinct hot spot where the highest percentiles are concentrated, and percentiles gradually decrease towards the northeastern direction on the map. The highest air toxics cancer risk percentiles (90 - 100 percentile) form a band of census tracts from west to east in the southern portion of the map, with several tracts overlapping the tracts covered by the toxic releases to air hotspot. Beyond the band of 90 - 100 percentile tracts for toxics cancer risk, percentiles drop to the 25 - 50 percentile range and reduce to 0 - 25 percentile in the northeast.

Due to the proximity and overlap of the highest percentiles for the two indicators, these maps imply that there may be a relationship between toxic releases to the air and air toxics cancer risk that would be worth investigating further.

While these visuals are thought-provoking, they only capture two indicators and a conclusive assessment cannot be made with this limited information. Further analysis of other air quality indicators and cancer risk factors in this region would be necessary to complete a detailed risk analysis.

Data Source

Environmental Protection Agency. *EJSCREEN: Environmental Justice Screening and Mapping Tool.* https://19january2021snapshot.epa.gov/ejscreen/download-ejscreen-data_.html

Citations

1. American Lung Association. *State of the Air 2026 Report*. https://www.lung.org/getmedia/32f0646d-c5de-4501-b0ac-07cd63c974d4/State-of-the-Air-2026-Report.pdf
2. Environmental Protection Agency. *AirToxScreen Frequent Questions*. https://www.epa.gov/AirToxScreen/airtoxscreen-frequent-questions
3. Environmental Protection Agency. *What is the Toxics Release Inventory?*. https://www.epa.gov/toxics-release-inventory-tri-program/what-toxics-release-inventory
4. San Bernardino County. *County Profile* https://indicators.sbcounty.gov/county-profile/
5. United States Census Bureau. *San Bernardino County, California* https://data.census.gov/profile/San_Bernardino_County,_California?g=050XX00US06071